

CareerSim: Gamification Design Leveraging LLMs For Career Development Reflection

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ABSTRACT

Graduates often struggle with challenges in their careers stemming from a lack of self-construction and decisive decision-making. The HCI community increasingly focuses on gamification as a means to foster reflection, with the emerging developments in LLM providing new opportunities for personalized gamified experiences. However, existing solutions fall short in terms of personalization and real-world data references. In response, our research introduces CareerSim, a role-playing game that leverages LLM's generation and reasoning capabilities. By integrating these with real-world based databases and game mechanics, we delve into gamification's role in reflective self-construction and decision-making. We aim to inspire designers to effectively incorporate LLMs in gamification design.

CCS CONCEPTS

• **Human-centered computing** → *Interaction design*.

KEYWORDS

Gamification design, Career development, Large Language Models

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1 INTRODUCTION

In the realm of career development, individuals construct themselves, impose direction on their vocational behavior, and make meaning of their careers [26]. In the increasingly globalized, competitive, and dynamic job market, understanding the relationship

between immediate decisions and long-term career prospects is a vital skill [9]. For university students, career development poses particular challenges, often exacerbated by a lack of relevant data references [28]. Moreover, the lack of understanding and reflection on self-construction and decision-making in career development poses a significant challenge. Apart from traditional avenues such as career mentoring [33, 35] and online career counseling [21], serious games and gamification have emerged as innovative and engaging approaches increasingly employed to support career development. For instance, SimCoach Games [22] and MeTycoon [9] utilize gamification elements to inspire youth in career exploration. Gamification design can be used to facilitate reflection [25, 32], but there is unresolved tension, such as oversimplifying the complexity of reality, encouraging players to perform actions that contribute to quantifiable performance rather than the act of reflection itself, and distracting the user from the core activity and its meaning [31].

The recent advancements in Large Language Models (LLMs) present an exciting avenue for personalized experiences and extended narrative arcs, opening up possibilities for a more nuanced approach. For example, the capabilities of knowledge extraction and text generation using LLMs can be utilized to construct an interactive AI-generated portrayal of a person's stories, attitudes, personality, and wisdom, enhancing user learning efficiency and motivation to acquire knowledge [19]. However, it faces challenges in covering professional knowledge related to career development and lacks real-world data as a reference [20]. There is a gap between the random content generated by LLMs and the gamification process used by designers because of their different capabilities and the dynamic and complex nature of career development. It is unclear how the LLMs will be used for gamification design work and the content and mechanics of creating the game.

Amidst this backdrop, our research addresses two research questions:

- RQ1: How does gamification design facilitate reflective self-construction and decision-making in career development?
- RQ2: What are the benefits and challenges of utilizing LLMs in gamification design, and how can they be effectively incorporated into the design process?

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This work presents our research-in-progress on expanding design possibilities of integrating LLMs in gamification design for career development. Towards this goal, we present a career development simulation RPG, CareerSim, incorporating two key elements: (1) Combining LLM’s personalized content generation with real-world data for game event development, enhancing real-world relevance and reflection. (2) Integrating LLM’s reasoning ability with career construction theory, employing gamification reward mechanisms to visually represent causality and attribute changes, stimulating players reflection on personal growth and decision-making. A preliminary user study demonstrates the effectiveness of CareerSim in fostering understanding and reflection on personal growth and career decisions. We hope that our research process brings new insights to other gamification designers.

2 RELATED WORK

2.1 Gamification in Career Development

Gamification, using game design principles and game elements in a non-game context [5, 7, 8, 14, 24], fosters personal connections and meaning in various fields, including business, education, health, and libraries [2, 6, 18]. Gamification can help players figure out their interests, preferences, and aptitude, contributing to stress-free decision-making, good academic performance, and increased satisfaction and confidence [1, 23]. Promoting reflection entails integrating processes of action and reflection, incorporating learning concepts and theories in the design of the gaming experience [4]. For instance, gamifying decision-making in dietary contexts stimulates reflection and positively influences behavior change [32]. In the realm of career development, the complexity and uniqueness of individuals’ journeys are addressed by the Career Construction Theory. This theory underscores the significance of integrating memories, current experiences, and future aspirations when making career choices [11]. Examples like SimCoach Games [22] and MeTycoon [9] utilize game elements to inspire youth in career exploration. However, existing research predominantly focuses on serious games for career guidance, emphasizing job applications, promotions, and career navigation [9]. Our work extends this by emphasizing causality and attribute dynamics, aiming to foster reflective experiences for career development.

2.2 Leveraging LLMs for Gamification Design

LLMs offer a unique opportunity for personalized narrative generation in gamification design. LLMs exhibit the ability to generate coherent and contextually relevant narratives, facilitating content creation for interactive storytelling [10, 19, 29]. Utilizing LLMs as generative agents enables personalized simulations, generating AI-driven portrayals of users’ stories, attitudes, and personalities [19]. Leveraging LLM-driven gaming can enhance player experience by granting players autonomy in open-ended conversations, potentially improving immersion and agency [3]. Moreover, research has also explored using LLMs to simulate challenging scenarios for job training [27], guiding self-monitoring of learning goals and reflection on job tasks [37]. Applying LLMs to real-world applications remains challenging due to their tendency to generate hallucinations and inability to use external knowledge [20, 27]. To address this, our proposed systematic approach involves constructing a

comprehensive database enriched with real-world information. By infusing game design with authentic data, we aim to bridge the gap between virtual and real worlds, creating a more immersive user experience and fostering meaningful reflection.

3 GAMIFICATION DESIGN

We refer to the DMC Pyramid [36] as our gamification design model, dividing the design process into three perspectives: Dynamics, Mechanisms, and Components (see Fig. 1). We first identify the dynamics, that is, the big picture of CareerSim, and then we consider the underlying mechanisms of the system. At last, we decide how to emulate the mechanisms in the form of components.

3.1 Game Genre

We chose the educational role-playing game as our game genre because it provides tools for reflection which keeps track of the player’s progress [17]. This specifically is in line with our needs for career reflection.

3.2 Dynamics Perspective

We set the main motivation of players of CareerSim as follows: A curiosity to see into the future and reflect on the present. The future is uncertain, and the professional image of the self is vague and unpredictable, but in CareerSim this uncertainty can be largely simulated. In the form of role-playing, players can live through the next 30 years to see the possibility of future development and construct their virtual occupational characters.

3.3 Mechanisms Perspective

In terms of game mechanisms, we first adopt the progression mechanism [15]. A player’s career development is divided into gradual stages, each with its themes, events and tasks. Players need to make decisions to progress through a certain stage and discover more career possibilities. We set program the game to end at retirement, which is also inline with our preliminary investigation. The theoretical support of this progressive design is Super’s vocational development theory [30].

As the player progresses, they can make a series of decisions and see the consequences of those decisions. This constitutes the mechanism of our chain of decisions and consequences. This is also consistent with the results we obtained in the previous investigation: 80 per cent of our target users consider "seeing the direct results of a series of career development decisions" their Number1 expectation in career simulation games. In the meantime, we believe that such a decision-making mechanism can encourage players to reflect on reality.

In addition, according to our interviews with targeted users, the simulation should be realistic and credible enough to "simulate real workplace as much as possible." We designed the opportunities mechanism to increase the uncertainty of the game and also to simulate the real career development environment according to Super. The opportunity occurs randomly, including workplace opportunities and events caused by external social, cultural, and economic factors.

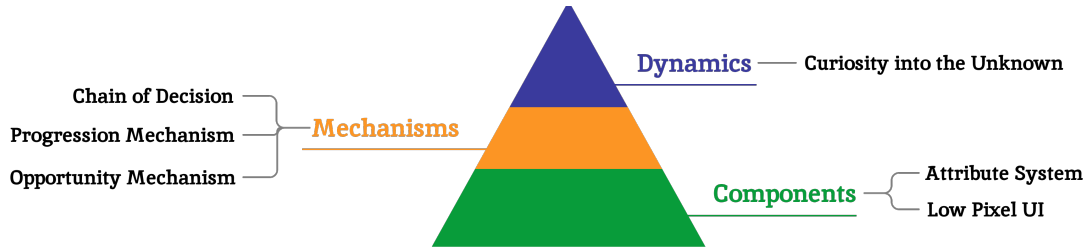


Figure 1: The design model of CareerSim

3.4 Components Perspective

To visualize the progression and consequences of decision-making, we incorporate the attribute system. The setting of the attribute is inspired by Holland’s vocational personality theory [12], including realistic, investigative, artistic, social, enterprising, and conventional attributes. At the beginning of the game, player defines the virtual character’s attributes to start the game. During the game, the attributes will change according to the player’s decision-making. At the same time, attributes also affect the type of events and the probability of success of their decisions. We also embraced low-pixel interface to enhance focus on simulated progress, aligning with RPGs’ narrative and game play emphasis, while ensuring broader accessibility for educational gamification.

4 GAME CONCEPT

In CareerSim, players take on the role of virtual characters, with various attributes set by the players at the initial stage. They experience a series of events that align with the possibility of career development and real experiences of alumni from a first-person vision, making decisions that shape their virtual career journey. Players can reflect on self-construction and decision-making in career development through reflection on each decision. CareerSim runs on the PC, and players can play it on their computers.

4.1 Game World

This year, an undergraduate student majoring in industrial design from X University has graduated. At the time, he needs to decide on his future path based on his preferences – whether to pursue further studies or enter the workforce. Events like this will shape his subsequent career development trajectory, requiring him to contemplate seriously and make careful choices within it. The player will continuously grow, with attributes and positions evolving throughout the decision made. When he turns 55, he will retire and receive a comprehensive report of his entire career life, he can review all the choices made in this report.

4.2 Game Play

Our game advances through events in the player’s career development. The overall game play can be divided into four main sections (see Fig.2).

(1) **Character Simulation.** At the beginning of the game, players need to define the character by editing the attribute system. The city attribute is input by the player, the initial value of the age attribute defaults to 22 years (upon undergraduate graduation),

and the characteristic attribute is scored by the player from 1 to 100 based on their situation. The system, combining the results of the characteristic attributes with Holland’s vocational personality theory[12], matches the player with the most suitable position. The attributes in the game are not fixed but dynamically adjusted based on the events encountered and choices made by the player.

(2) **Career Development Events.** After the game begins, players will embody a virtual character and go through career development events generated by the system. Each event consists of characters, time, content, and multiple interactive options for players to choose from. The generation of career events is influenced by the player’s career development stage (determined by the age phase, referencing Super’s vocational development theory [30]) and personal attributes, as will be discussed in Section 5.3.

(3) **Decision-making and Impact.** Players draw upon their real-life experiences to make decisions from the multiple options provided in the game, and the decisions will influence the development of their virtual career journey. When making decisions, players need to consider the potential consequences of each option and make the decisions carefully. After each decision, the character’s attributes change based on the player’s decision. The character’s age is also randomly updated to a value within the age range of the next career development stage. These changes will influence the generation and matching of subsequent events.

(4) **Post-game Review.** The game will generate a comprehensive career development report for the player. This report aims to assist players in thoroughly reviewing significant events encountered and evaluating their entire career journey. By reading the report, players can retrospectively explore the relationship between events and the overall trajectory from a third-person perspective, understanding the impact of each decision on the outcomes, and thus facilitating reflective insights.

5 IMPLEMENTATION

The implementation of CareerSim can be divided into four main sections (see Fig.4).

5.1 Input

Input comprises two aspects:

(1) **Players input their attributes in the terminal to initiate the gameplay.** We interpret the initial player attributes as triggers for the first set of events generation. While the game proceeding, the system can dynamically change the attributes based on players’ career event decisions.

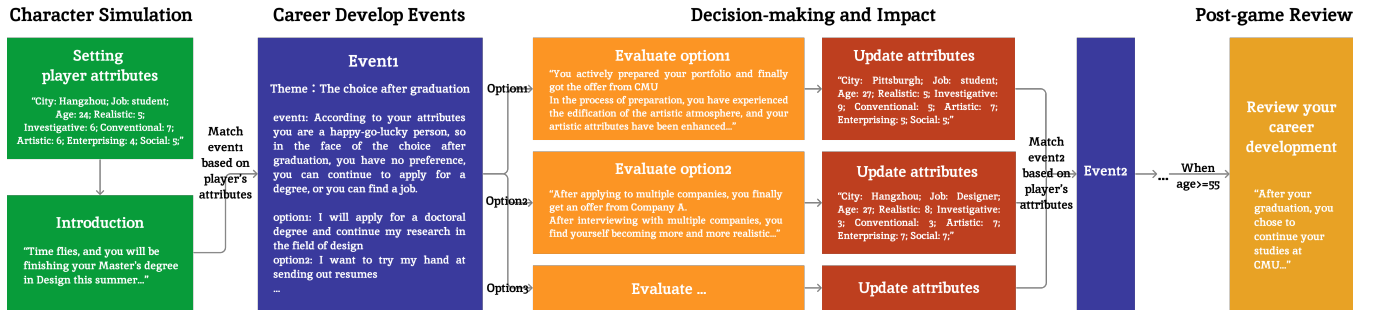


Figure 2: Four main sections of CareerSim

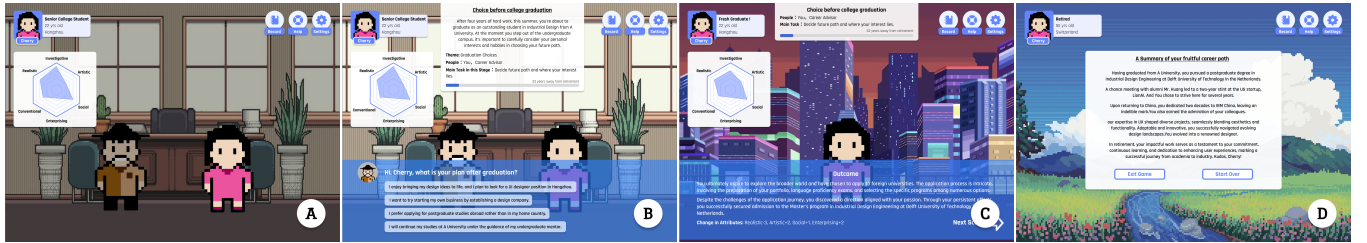


Figure 3: The user interface of CareerSim. Upon entering the game, players need to set attributes to customize their characters (A). Once the game begins, players will encounter career development events based on their attributes, requiring them to make choices (B). After each event, the player's attribute values will change based on their choices (C). At the end of the game, a career development report will be generated to help users review their journey (D).

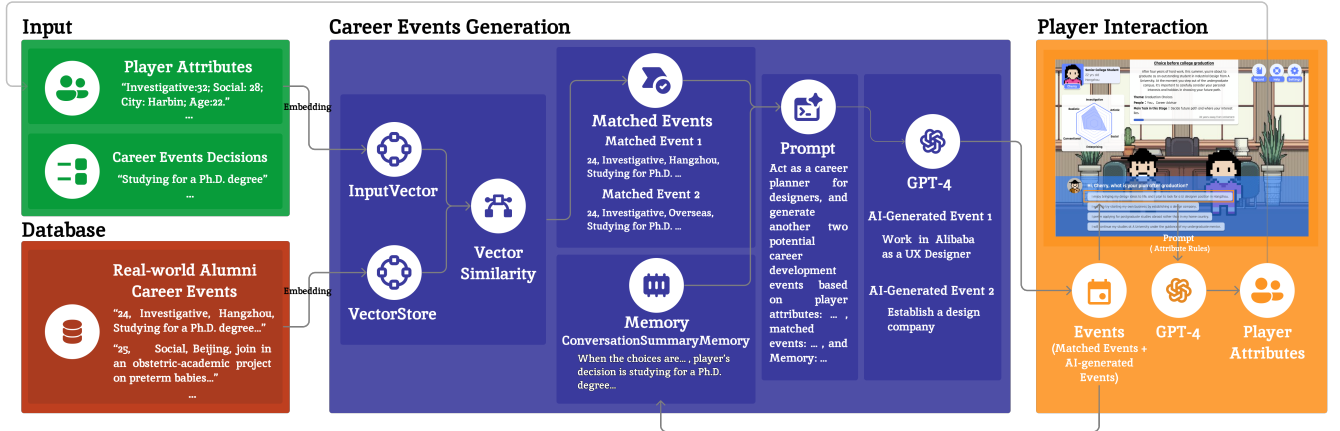


Figure 4: The architecture of CareerSim

(2) **Players make career events decisions, propelling the progression of the game.** The main player interaction of CareerSim is to make decisions on virtual career events, as detailed in Section 5.3. These decisions directly determine the outcomes of current career events in the in-game world and influence the next set of event generations.

5.2 Database

The database is composed of real-world alumni career development events, providing relevant data references for CareerSim. Our first-hand data is sourced from official reports on alumni from the author's university platform, ensuring the authenticity of the data. These reports, including text and video interviews, span over two decades and cover the career events of 50 outstanding graduates from the classes of 1990 to 2013. According to Super's vocational

development theory [30], we organized alumni career events based on the stages of occupational growth.

Subsequently, experts from the university's career development centre were invited to categorize these events based on the six attributes adhering to Holland's vocational personality theory [12], providing qualitative processing of the events. Two members of the research team then annotated these events in a structured format, including career development stage, attributes, and a succinct paragraph natural language description of the events to ensure the granularity of the data for efficient matching and retrieval.

5.3 Career Events Generation

We employed Langchain and large language model(GPT-4) to generate career development events. This approach can stimulate players to reflect on career development, by simulating career experiences authentically and individually.

Langchain is a framework for developing applications powered by language models, with practical modules including Retrieval, Model I/O, Chains and Memory [34]. To generate the real-world matched events, we use the Retrieval module which splits and embeds the career events in the database into 256-dimensional dense vector chunks, ensuring a complete description of each alumni event. By comparing cosine similarity, we can perform a semantic search and find the chunks containing the career events matched to player input. This process yields matched career events. Additionally, we used the Memory module, applying the ConversationSummaryMemory component, to summary and memorize the players' decisions.

For the generation of random events, we employed LLM GPT-4. The Model I/O module in Langchain serves as an interface to connect with GPT-4. The matched career events retrieved from the database, decisions memory, together with player inputs, are conveyed into a predefined prompt template to generate prompts. Based on the prompt, GPT-4 generates two or three AI-generated career events. Eventually, players are presented with four to five career development options, with two to three matched events retrieved from the database, and several AI-generated random events. Players can make decisions by selecting from these options, thereby experiencing the uncertainty and possibilities inherent in professional career trajectories.

5.4 Player Interaction

5.4.1 Dynamic Feedback. The gamification design incorporates a dynamics perspective, enabling real-time interaction between player actions and the game environment. Dynamic feedback is primarily manifested in four aspects:

(1) The interactive process consists of players inputting initial attributes and players making career event decisions. Players input initial attributes to start the game. Based on the initial attributes, the system extracts matched events from the database and guide GPT-4 to generate random events. Therefore, players' first set of career options includes both matched events and AI-generated events. After players make career event decisions, the system calls upon GPT-4 again to update player attributes based on a predefined prompt following Holland's vocational personality theory. The

updated player attributes and player's decisions in the first set generate the next set of career options.

(2) The generation of subsequent events is influenced by the players' decisions in the previous set. Players' decisions in the previous set influence players' input, which determine the matched events after vector similarity comparison. In addition, the players' decisions are summarized and memorized in the Memory module, which together with the matched events, and player attributes, generate prompts to influence AI-generated events.

(3) Unity is utilized to connect the system with the player interface. In terms of interface design, various working scenes such as companies, university labs, and art galleries are predefined. Random scenes like cities are also designed to showcase unplanned random events. Virtual characters are categorized into male and female genders, with various occupational characters such as corporate employees, researchers, and artists. Based on Super's vocational development theory [30], each character is designed with age images corresponding to the five stages. As players make different career decisions, the game scenes and virtual character images evolve accordingly.

(4) When a player's age attribute is up to 55, the system defaults the player to the retirement stage. The system retrieves all career development event decisions from the Memory module and utilizes GPT-4 to generate a personalized career development report for the player.

5.4.2 Evaluation of Player Interaction. LLMs are known to generate hallucinatory content, which is defined as content that is "nonsensical to the provided source content" [13]. CareerSim generates events with continuity and relevance, enhancing the player's career simulation experience. To achieve this, we establish prompts that leverage GPT-4 to process player attributes. Combined with player decisions, these inputs facilitate the retrieval of similar alumni data, ensuring contextual relevance. Additionally, the Memory module is applied to remember player interactions. This memory influences subsequent interactions, ensuring the coherence of the output of career development events.

6 PRELIMINARY USER STUDY

In our preliminary user study, we recruited twelve college students aged between 18 and 24 to experience CareerSim and undergo semi-structured interviews. From our initial observations, we found that CareerSim creates immersive lifelike experiences with multiple decision pathways.

First, users actively integrated their real-life situations into the game's attribute settings and decision-making. Initial attribute settings often mirrored users' actual circumstances rather than diverging significantly. This suggests that the system design intuitively inspires users to conduct personalized career exploration. Knowing that the game utilized an alumni database to generate events, users expressed interest in exploring alternative choices after initial decisions, aiming to observe how alumni approached similar situations. For instance, User 5 remarked, *"Feels very realistic, providing an objective understanding of future scenarios and allowing for early experiences and preparations."* This shows that apart from completely simulating their own situation, users are inspired to explore more career possibilities. However, the sense of immersion and relevance

of games may decrease with age. User 7 mentioned, "When I just graduated and started working, I had a clear plan for my choices, but when I reached 40, I didn't know what to consider for decisions anymore."

Second, in causal relationships and attribute changes, We found that changes in visual attributes can indeed lead to reflection, just as we envisioned in the design of game mechanics. To our surprise, cruelty or failure is more likely to promote reflection, specifically referring to decisions that do not meet expectations and a decrease in attribute values. For instance, User 1 expressed, "For 'me,' I don't usually think about the pitfalls in career development. Therefore, facing failures and setbacks in the game makes me reflect on my choices more easily." In this case, the virtual "adversity" created through changes in events and attribute values may enhance the user's reflection and resilience in reality[16].

In conclusion, our preliminary study demonstrates the effectiveness of the system's design and its gamified design in triggering career reflection. However, challenges emerge due to limited contextual depth and difficulties in offering nuanced responses, especially in scenarios projecting into the distant future. This suggests that future versions of CareerSim should introduce richer themes in career development stages, such as family factors.

7 CONCLUSION AND FUTURE WORK

In conclusion, we report on a design process utilizing LLM for gamification design in career development simulation, aiming to facilitate players' reflection on self-construction and decision-making. Through initial observations, we identified that leveraging LLMs for gamification design offers motivating benefits, providing appropriate feedback for diverse explorations. Future improvements focus on (1) integrating real-time internet databases for up-to-date employment scenarios and (2) refining LLM with career development knowledge for more realistic causal relationships. Our evaluation plan involves surveys, interviews, and scales to gauge user experience, decision-making, and self-construction understanding. Through iterative versions, we aim to explore gamification's potential across learning domains, contributing to LLM-based gamification theory.

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